

Frederick Searancke

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JOB EXPERIENCE

Samarajames — Bespoke Software Developer

June 2024 - Aug 2024

Produced AI live chat application integrated into the existing live chat software. Created a custom LLM model trained on historical employee live chats with customers to assist sales employees.

Implemented a RAG (database system) covering 698 products and 3000+ vector embeddings, enabling fast real-time access to relevant product information. This interfaces with the LLM to feed up-to-date relevant information. Responses are fast enough for communication in real time.

PROJECTS

Trading Analysis — *backtesting stock market trading strategies*

Run detailed backtests for specialised retail trading hypotheses. Used software QuantConnect as a backtesting framework, and for backtest permutations through an API.

Wrote Python code to optimise strategy parameters within a given hypothesis using in-sample and out-of-sample testing. Run backtests on equity indexes such as SPX, and commodities futures such as crude oil, and futures options on commodities like corn, wheat, and crude oil.

Tested entry signals using hypothesis testing, obtaining statistically significant results ($p=0.006$), and visualised findings for presentation.

CURRENT RESEARCH FOCUS — *Volatility Risk Premium (VRP)*

Evaluating options VRP ideas and portfolio interaction effects (correlations, drawdowns, diversification) using the same backtesting + significance-testing framework.

EDUCATION

University of Warwick — *Computer Science First-class (expected)*

Sept 2024 - June 2027

A Levels — A*AAA (Maths, Further Maths, Physics, Computer Science)

Sept 2022 - June 2024

SKILLS

Python (11 years)
Market Backtesting
Strategy Development
Python for data analysis
Python for APIs
Python for Automation

AWARDS/ACHIEVEMENTS

Cambridge Computer Science admissions assessment — passed 2023

Built automated Python→Blender pipeline to generate client-ready 3D models (paid work)

Built a Gmail API pipeline that turned unstructured data into structured JSON data

LANGUAGES

Python (11 years), Java

PERSONAL INTERESTS

Fencing, Skiing, Boxing,
BASI level 1 ski instructor,
Gold Duke of Edinburgh

